Learning Stop/Continue Strategies in *Can't Stop*

CS 441 Final Project — Fall 2026
Group members: *[add names and net IDs]*
Code: [repository link to be added]
Data and experiment artifacts: [repository link to be added; see `cant_stop_baselines/runs/`]
Video: *[add 5–6 minute video link]*

Abstract

We study whether machine learning can improve the strategic Stop/Continue decision in the two-player board game *Can't Stop*. The application observes a game state after a successful roll and chooses either to bank the temporary progress or risk another roll. We compare a rule-based baseline, a supervised linear model, Double DQN, and PPO. Dice-pair selection is held fixed to the Glenn–Aloi constant-weight policy for the main comparison, so the experiment isolates the Stop/Continue policy.

The strongest retained model is a PPO policy trained from scratch with domain features and a 60% Monte Carlo / 40% smart Rule-of-28 opponent mixture. On a fresh common evaluation set of 3,000 games per matchup, it won 64.6% against Rule28-smart and 55.4% against the approximate MC agent. It significantly outperformed the retained DDQN reference by 3.2 percentage points and 2.7 points, respectively, under paired bootstrap intervals. Two additional seeds support the PPO result: PPO averaged 64.25% against Rule28-smart and 55.35% against MC, compared with 59.50% and 50.40% for DDQN. However, PPO and the multi-policy Ridge model were close in direct head-to-head games, so we do not claim universal superiority. Learned dice pairing and several fine-tuning and supervised-DNN variants were negative results.

1. Problem Description and Motivation

1.1 Application

The proof-of-concept application is an automated *Can't Stop* player. Given the board position, temporary runners, banked progress, the opponent's progress, and the result of the current dice pairing, the player decides whether to:

- **Continue:** roll again and potentially gain more progress or bust; or
- **Stop:** bank the current temporary progress and pass the turn.

The game is a useful decision-making problem because every decision trades off short-term risk against progress toward claiming three columns. A simple heuristic can be reasonable, but the value of continuing depends on the whole board, opponent threats, completed columns, and the probability of busting.

1.2 Intended user and use case

The intended user is a person playing or studying *Can't Stop*. The application can act as an opponent, provide a recommendation during a human game, or serve as an educational tool that explains why a state is risky. The practical benefit is a reproducible strategy assistant rather than a claim that the game has one universally optimal move.

1.3 ML formulation

This is a two-action sequential decision problem. At each eligible decision state, the input is a fixed-length vector describing the current player's state. The output is a policy over `{Continue, Stop}`. The principal reward is terminal: `+1` for a win, `-1` for a loss, and `0` otherwise. The main PPO and DDQN experiments use fixed GA dice pairing, while the learned-pairing experiments additionally predict a legal pairing action.

The game engine is independent of the policies. It enforces the standard board heights `(3,5,7,9,11,13,11,9,7,5,3)`, at most three active columns, temporary progress loss on a bust, and victory after three claimed columns. The implementation was checked by a regression suite and seeded reproducibility tests.

2. ML Approach

2.1 Baselines

We used several increasingly strong baselines:

1. **Rule28-smart:** a transparent heuristic based on Rule of 28 and Glenn–Aloi constant move weights. It stops at a score threshold or when a temporary runner reaches a column top.
2. **Approximate MC:** fixed GA pairing plus one-turn Monte Carlo rollouts, with Rule28-smart used after simulated actions. It is an approximate remainder-of-turn policy, not full-game MCTS.
3. **Ridge-domain:** a supervised Ridge regressor predicting the approximate margin `Q(Continue) - Q(Stop)` from domain features. It continues when the predicted margin is positive.
4. **MC-counterfactual DDQN:** the retained RL reference, trained with Double DQN, five-step returns, domain features, and counterfactual MC supervision.

The final report focuses on PPO, DDQN, and Ridge because Random Forest and the single-source and multi-policy DNNs were clearly weaker in complete-game evaluation.

2.2 PPO model

The promoted PPO policy uses a 68-value domain observation and a 128x128 MLP. It was trained for 100,000 episodes with terminal-only reward, 512-step rollouts, four PPO epochs, minibatches of 256, learning rate `1e-4`, and fixed GA pairing. The training opponent mixture was 60% MC and 40% Rule28-smart. Training MC used 16 rollouts and a future-roll limit of 20; held-out evaluation used MC-64. The promoted checkpoint is the best validation checkpoint from seed 451, episode 95,021:

```
cant_stop_baselines/runs/ppo_mc_focused_seed451/top_1.pt
```

Checkpoint selection maximized the minimum validation win rate across the two primary opponents. We retained the checkpoint rather than the final training weights because validation performance oscillated substantially during PPO training.

2.3 DDQN model

The DDQN reference uses a 128x128 network, 68 domain features, five-step returns, fixed GA pairing, terminal game rewards, and an opponent mixture of Rule28-smart, MC, pure GA Rule 28,

classic Rule 28, and random agents. The retained checkpoint is:

`cant_stop_baselines/runs/ddqn_mc_counterfactual_seed447/best.pt`

This model is the main RL baseline because it was selected before the final PPO comparison and remained more reliable than attempted self-play and fine-tuning variants.

2.4 Supervised Ridge model

We collected 10,000 decision states from simulator-generated games. For each state, the pipeline forced Stop and Continue using shared future random seeds and estimated each action's full-game win probability with paired rollouts. The target was the resulting action margin. A grouped 70/15/15 split by source game prevented states from the same game appearing in multiple splits. The multi-policy Ridge model was trained on states drawn from five policies (Rule28-smart, MC, Ridge, PPO, and DDQN), using the same counterfactual label procedure. This distribution improved Ridge robustness over the original single-policy dataset.

3. Data Collection and Curation

All data were generated by our own simulator rather than taken from Kaggle, benchmark datasets, or a pre-curated ML dataset. Each generated game records the seed, seat assignment, winner, turns, rolls, busts, and stops. Decision states are stored with fixed-length features and metadata identifying the source game and policy.

The state representation contains 58 raw values. The domain representation contains 68 values, including active/top marker counts, immediate completion and win indicators, board-progress lead, risk-adjusted turn progress, normalized Rule-of-28 score, and opponent threat. The strategic ablation adds per-column temporary progress and completion-threat features.

For supervised labels, the same state and future random seeds were used for the forced Stop and forced Continue branches. This reduces label noise caused by unrelated dice variation. Labels remain approximate because the rollout policy after the forced action is Rule28-smart; they are not optimal-game ground truth. Source games, rather than individual states, were split before training. Evaluation games were generated with fresh seeds that were not used for collection, training, or checkpoint selection.

For every formal model comparison, seats were alternated. The two games in a seat-swapped pair share a dice seed, so paired bootstrap is more appropriate than treating all game outcomes as independent. We report 5,000 paired bootstrap samples and interpret intervals as descriptive paired inference.

4. Experimental Design

4.1 Primary evaluation

The final common evaluation used seed `1000454`, 3,000 games per matchup, alternating seats, MC-64, and 5,000 paired-bootstrap samples. All models faced the same fresh dice seeds and the same two primary opponents.

4.2 Primary results

Model	Rule28-smart	MC	Minimum
PPO-451 + GA	**64.6%**	**55.4%**	**55.4%**
DDQN reference	61.4%	52.7%	52.7%
Ridge-domain	60.1%	51.1%	51.1%

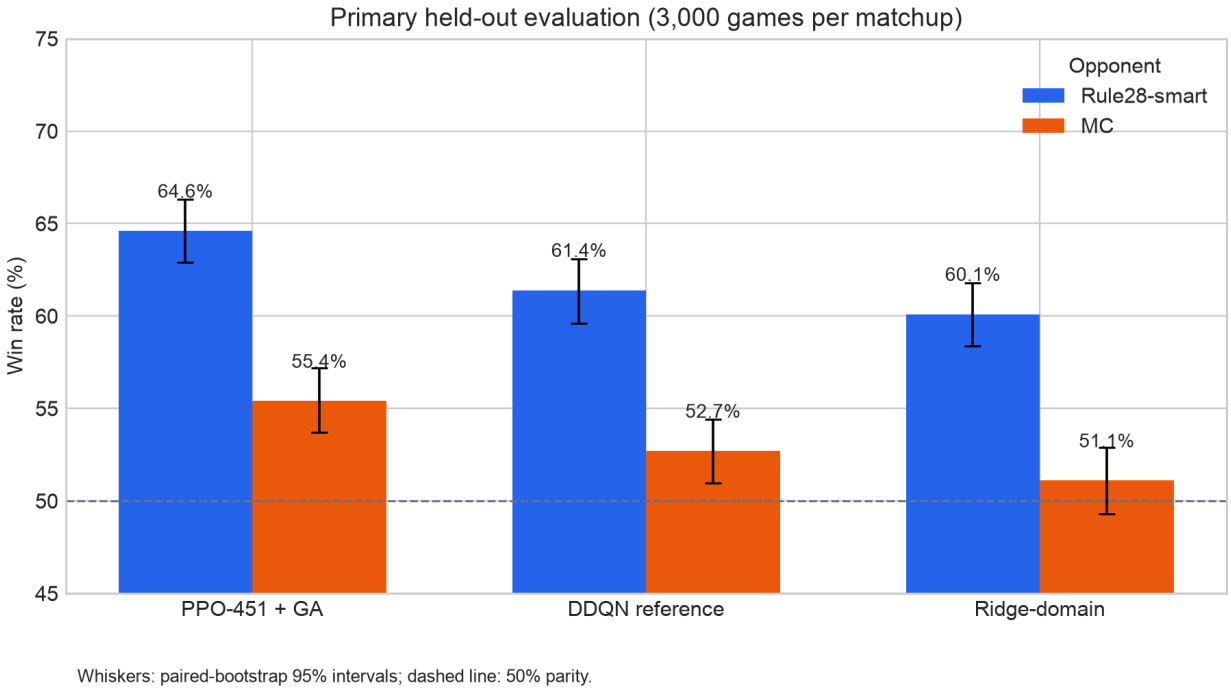


Figure 1. PPO has the highest point estimate against both primary opponents, and its paired intervals are above the DDQN and Ridge comparisons reported below.

PPO's paired advantage over DDQN was **+3.2 points** against Rule28-smart, with a 95% interval of **+1.2 to +5.3**, and **+2.7 points** against MC, with an interval of **+0.6 to +5.0**. PPO's advantage over Ridge was +4.5 points against Rule28-smart (interval +2.2 to +6.7) and +4.3 points against MC (interval +1.9 to +6.6). DDQN versus Ridge was not statistically resolved.

The primary PPO results were not merely wins against weak opponents. In a separate 3,000-game external-opponent evaluation, PPO won 80.9% against the released Glenn–Aloi agent and 65.8% against the Couto-GA-stop adaptation. The corresponding PPO rates against Rule28-smart and MC were 64.1% and 55.3%. These results also show opponent dependence: Ridge slightly exceeded PPO against Glenn–Aloi (81.7% versus 80.9%) in that evaluation.

4.3 Additional-seed robustness

We retrained each retained family with two additional seeds and evaluated on a common fresh set of 3,000 games per matchup (seed `20261101`). The values below are means over the two new training seeds, not pooled games and not a claim that every future seed will match them.

Model family	Rule28-smart mean	MC mean	External-opponent mean	Minimum mean
PPO (seeds 460/461)	**64.25%**	55.35%	74.6%	55.35%

Model family	Rule28-smart mean	MC mean	External-opponent mean	Minimum mean
DDQN (seeds 448/449)	59.50%	50.40%	71.0%	50.40%
Ridge (seeds 642/643)	63.25%	**55.55%**	**74.9%**	**55.55%**

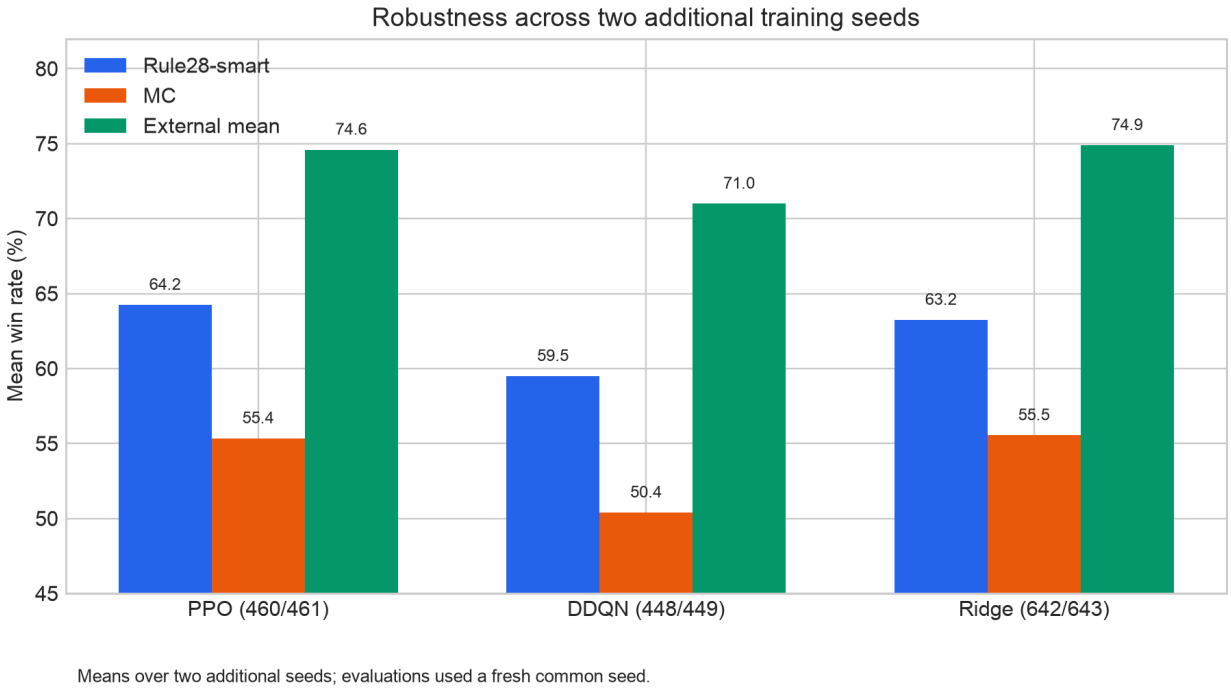


Figure 2. The additional-seed check shows that PPO and Ridge are close, while DDQN has the lowest mean and greater seed sensitivity.

PPO was consistent across the two additional seeds. DDQN was more seed sensitive, while Ridge was also stable and close to PPO. Thus the evidence supports PPO as the strongest promoted RL policy, but not as a uniformly dominant policy family.

4.4 Direct head-to-head results

The six additional checkpoints were also compared directly in 3,000-game pairings using seed `20261102`. Averaging the two cross-family results gives:

Pair	Approximate first-model win rate
PPO versus DDQN	56.1%
PPO versus Ridge	49.2%
Ridge versus DDQN	56.3%

The direct results qualify the primary-opponent conclusion. Performance is non-transitive and depends on the opponent; PPO is clearly strong against the selected external baselines, but the new Ridge checkpoints were competitive and slightly favored in direct games.

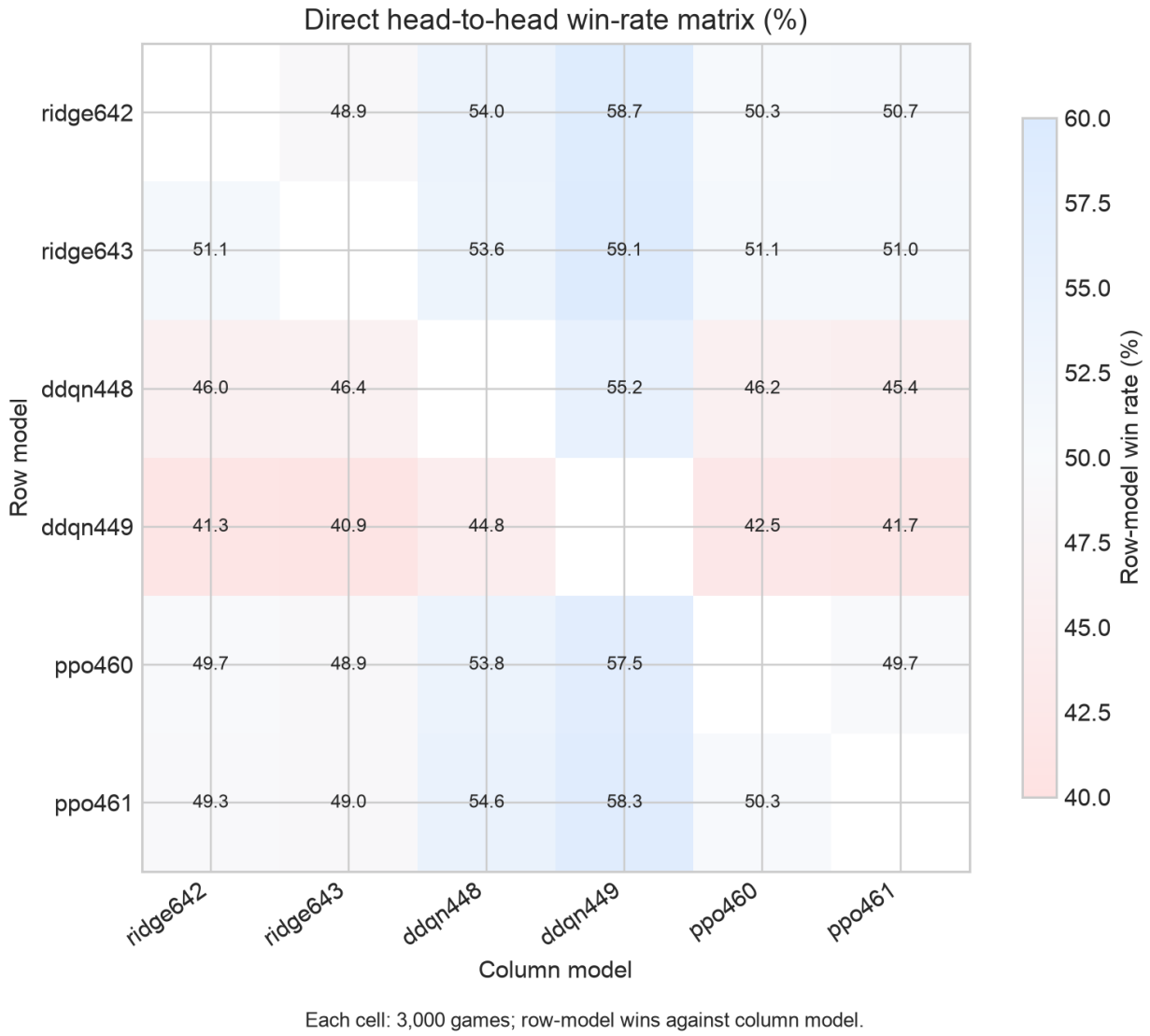


Figure 3. Rows are the models whose wins are counted; columns are their opponents. The matrix makes the non-transitive relationship visible instead of reducing it to one overall ranking.

4.5 Ablations and negative results

Negative results were retained as part of the model-selection evidence:

- A learned-pairing PPO run destroyed performance when all parameters were updated. A pairing-only diagnostic preserved the Stop/Continue policy but still scored only 50.4% versus Rule28-smart and 44.5% versus MC, below PPO (64.6% and 55.4%) and DDQN (61.4% and 52.8%).
- A supervised DNN trained on the original counterfactual dataset scored 42.7% and 36.2% against Rule28-smart and MC, despite 68.2% validation action accuracy. The multi-policy DNN was weaker still at 32.3% and 28.7%.
- Multi-policy Ridge improved over the original Ridge (62.7%/57.4% versus 59.4%/53.2% in its comparison), showing that state-distribution curation helped the linear model but did not

make the DNN effective.

- Self-play, conservative fine-tuning, targeted fine-tuning, reference-replay fine-tuning, and a counterfactual PPO screen all regressed on held-out games. For example, the reference-replay DDQN fell to 58.6%/50.4%/46.7% against Rule28-smart/MC/Ridge.

These ablations prevent us from attributing all improvement to neural-network capacity or to additional training alone.

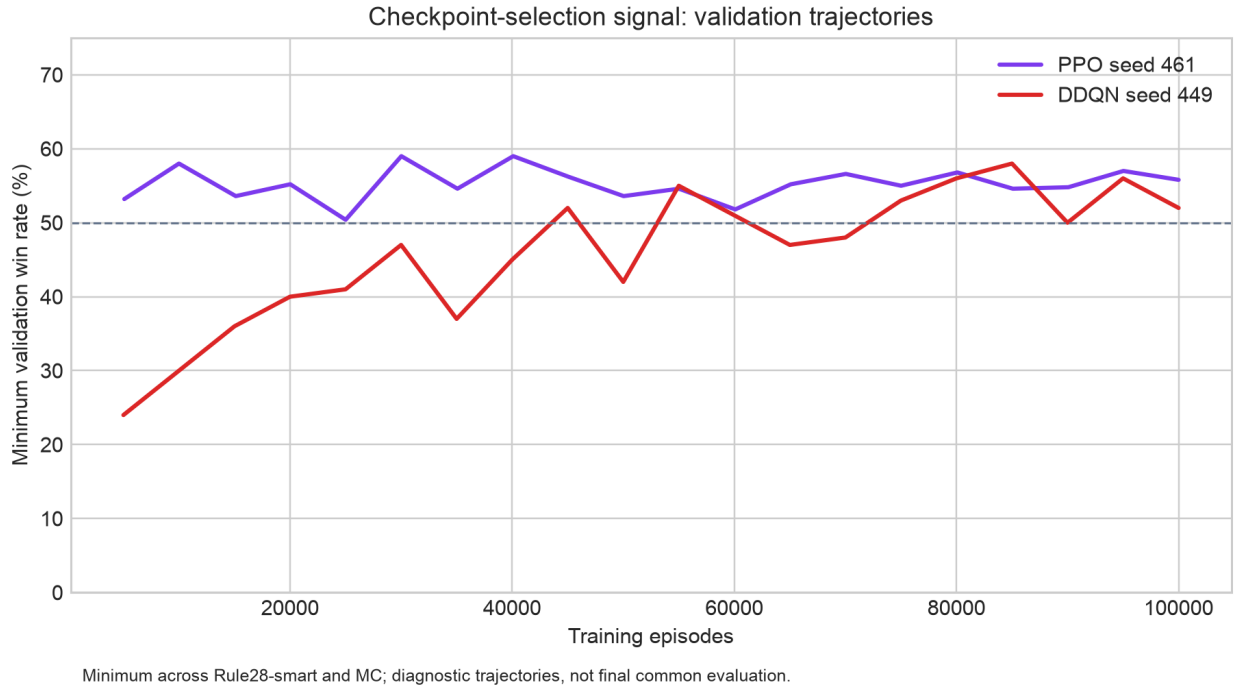


Figure 4. Validation performance is noisy and checkpoint-sensitive. This is why the final selection used a held-out common evaluation rather than the last training checkpoint.

5. Discussion

The answer to our research question is **qualified yes**. PPO learned a stronger Stop/Continue policy than the retained DDQN and Ridge baselines when tested against the primary heuristic and approximate MC opponents. The advantage was statistically supported in the paired common evaluation and was largely reproduced by two additional PPO training seeds.

The answer is not "PPO always wins." First, direct head-to-head results are near 50–50 against Ridge and show non-transitive strategy interactions. Second, the paired intervals are based on shared dice-seed pairs and should not be interpreted as independent-game confidence intervals. Third, training and evaluation use simulator-generated games, so the conclusions concern this implementation of two-player *Can't Stop*, fixed GA pairing, and the selected opponent distribution.

The most useful engineering result is that selecting a stable checkpoint and curating state distributions mattered more than adding model complexity. Learned pairing was unsuccessful under the current terminal-reward setup, while multi-policy data improved Ridge. The final application should therefore ship PPO-451 + GA as the primary player, keep DDQN as an RL reference, and offer Ridge as a lightweight alternative.

6. Limitations and Future Work

1. The main study uses two players and fixed GA dice pairing; a learned joint pairing policy was tested but not solved.
2. MC is an approximate one-turn rollout agent, not full-game MCTS or an optimal oracle.
3. The supervised labels are approximate Rule28-smart rollouts, so offline accuracy is not a substitute for complete-game evaluation.
4. PPO and Ridge have only two additional retraining seeds in the final robustness check. More independent seeds would strengthen generalization claims.
5. The current proof of concept has no graphical interface and does not claim support for three- or four-player variants.

Future work should investigate legal-action-masked pairing with a better training objective, broader independent-seed evaluation, and calibrated explanations of the predicted risk of continuing.

7. Reproducibility

The source tree contains the simulator, agents, training scripts, tests, model checkpoints, raw per-game CSV files, JSON summaries, and configuration files. The most important artifacts are:

- `cant_stop_baselines/docs/EXPERIMENT_STATUS.md` — experiment log and final decisions;
- `cant_stop_baselines/runs/final_common_eval_seed1000454/` — primary common evaluation and raw games;
- `cant_stop_baselines/runs/final_retrain_head_to_head_seed20261102/` — additional-seed direct comparisons;
- `cant_stop_baselines/runs/ppo_mc_focused_seed451/top_1.pt` — promoted PPO;
- `cant_stop_baselines/runs/ddqn_mc_counterfactual_seed447/best.pt` — DDQN reference;
- `cant_stop_baselines/runs/ridge_multi_policy_seed642/` and `seed643/` — additional Ridge models.

From `cant_stop_baselines/`, the regression suite can be run with:

```
python3 -m unittest discover -s tests -t . -v
```

8. Acknowledgements and Contributions

We acknowledge the authors of the Rule of 28, Glenn–Aloi heuristic, and Couto/Sketch-SA(O) work cited below. Their methods are used as documented baselines or adaptations; they are not presented as our original algorithms.

Contributions: *[replace this paragraph with each member's concrete work, for example simulator/rules, data pipeline, PPO/DDQN training, evaluation, analysis, and report/video preparation.]*

References

1. Martin Keller. “Can't Stop? Try The Rule of 28.” 1986.
2. James Glenn and Christian Aloi. “A Generalized Heuristic for Can't Stop.” FLAIRS, 2009. https://cdn.aaai.org/ocs/123/123-2430-1-PB.pdf
3. Leandro C. Medeiros, David S. Aleixo, and Levi H. S. Levis. “What Can We Learn Even From the Weakest? Learning Sketches for Programmatic Strategies.” AAAI, 2022. https://doi.org/10.1609/aaai.v36i7.20744

4. J. Cameron Carter. “Using Machine Learning to Optimize the Board Game Can't Stop.” 2026, user-provided thesis.